Chapter 10. Deflection of statically indeterminate beam

10.1. Introduction

In finding the shear force and bending moment in a beam, we encountered problem such as a cantilever beam AB shown in Fig. 1(a). The left end A of the beam is fixed and there is a uniform distribution force q applied over the beam.

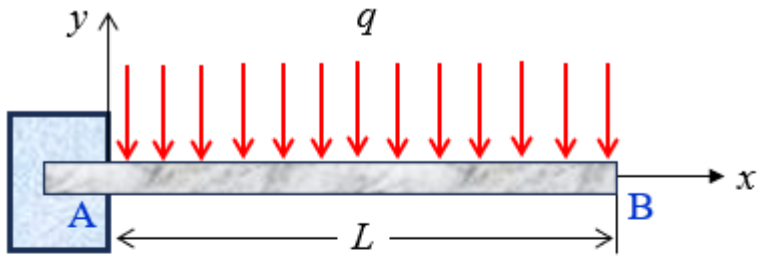


Fig. 1(a)

To find the internal shear force (V) and bending moment (M) in this beam, we drew a free-body diagram (Fig. 1(b)). We used a free body diagram on the right-hand side of the cut point. And we used the equilibrium equations:

$$\sum M_x = 0, \qquad -q(L-x)\frac{L-x}{2} - M = 0, \qquad M = -\frac{q(L-x)^2}{2} \tag{a}$$

$$\uparrow \sum F = 0, \qquad -q(L-x) + V = 0, \qquad V = q(L-x) \tag{b}$$

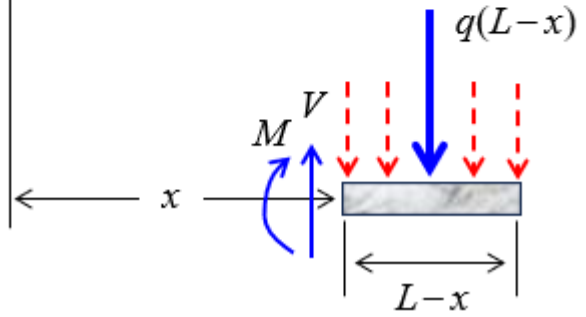


Fig. 1(b)

We saw the number of unknowns (M, V) equals the number of available equilibrium equations. The internal shear force and bending moments could be determined solely based on the equilibrium equations. The beam is *Statically Determinate*.

We now look at the problem of Fig. 2(a). There is a roller support at the right end of the beam.

We again draw a free-body diagram (Fig. 2(b)). The roller at the right end provides the vertical support to the beam. Therefore, there is an external reaction by the roller (R) in vertical direction as shown in the free-body diagram of Fig. 2b.

The equilibrium equations for the free-body are:

$$\sum M_x = 0, \qquad -q(L-x)\frac{L-x}{2} - M + R(L-x) = 0 \qquad \text{(c)}$$

$$\uparrow \sum F = 0, \qquad -q(L-x) + V + R = 0, \qquad \text{(d)}$$

Equations (c) and (d) are two equations in three unknowns (M, V, R) and therefore, they can not be solved. So this beam is *Statically Indeterminate*.

Because the internal bending moment can not be obtained directly. More information is needed to determine the deflection of statically indeterminate beam.

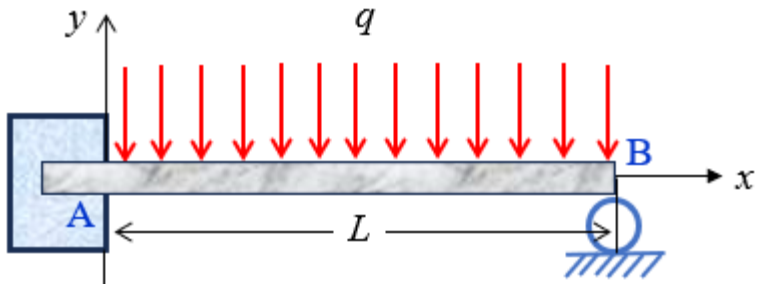


Fig. 2(a)

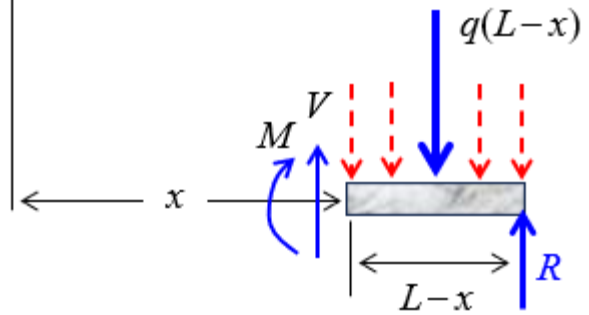
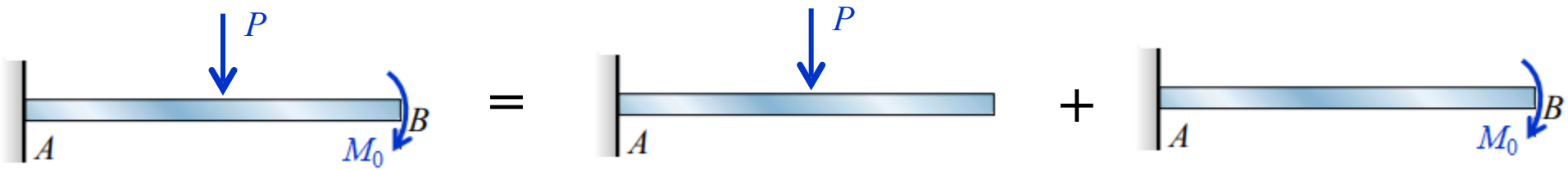


Fig. 2(b)

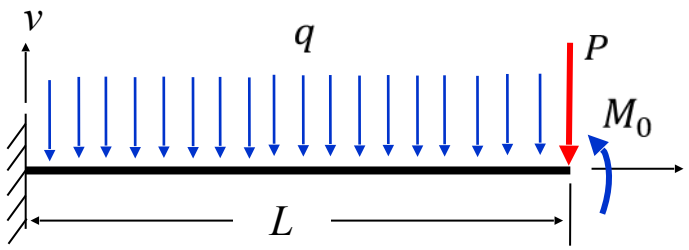
10.2. Obtaining deflection by superposition method

Assume the material of beam is linear and elastic, and the deformation of the beam is small.

The deflection and slope of a beam that simultaneously supports multiple loads can be obtained by linear superposition (i.e., by addition) of the effects of the loads acting separately.



The deflection of a statically determinate beam can be easily obtained by superposition. We take below beam as an example.



For this problem, there are three subproblems:

1. A cantilever beam with a uniform distribution force q.

2. A cantilever beam with the application of a concentrated (point) force P at its free end.

3. A cantilever beam having a couple (moment) at its free end.

In Chapter 8 (or from the table of textbook), we know the deflections curves for the beams of these subproblem. These are:

$$v = \frac{q}{24EI}(-x^4 + 4Lx^3 - 6L^2x^2) \qquad \text{(a)}$$

$$v = \frac{P}{6EI}(x^3 - 3Lx^2) \qquad \text{(b)}$$

$$v = \frac{M_0}{2EI}x^2 \qquad \text{(c)}$$

The deflection of the beam under these combined forces are obtained by adding Eqs. (a), (b) and (c) to get:

$$v = \frac{q}{24EI}(-x^4 + 4Lx^3 - 6L^2x^2) + \frac{P}{6EI}(x^3 - 3Lx^2) + \frac{M_0}{2EI}x^2$$

Example . Solve for the reactions on the beam shown in Fig. 1. Using information on cantilever beams from Table.

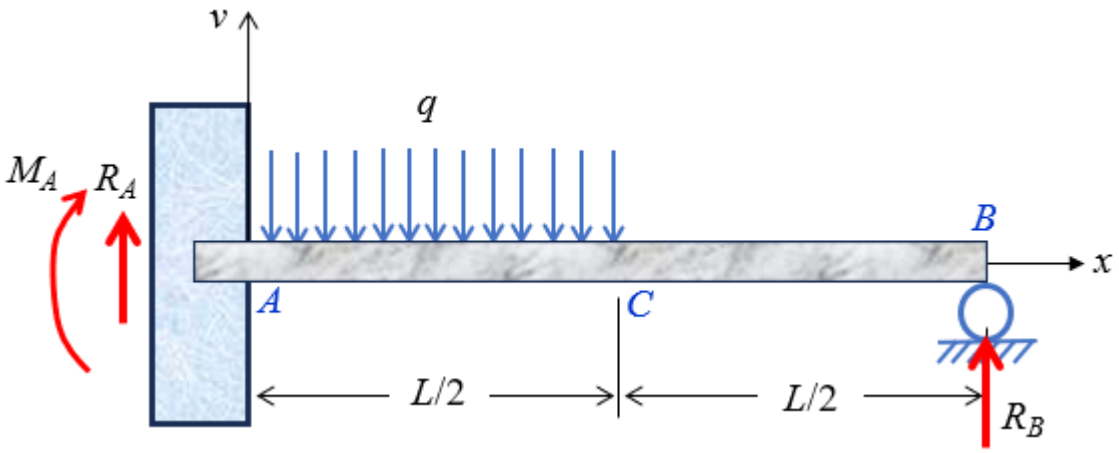


Fig. 1. Original problem

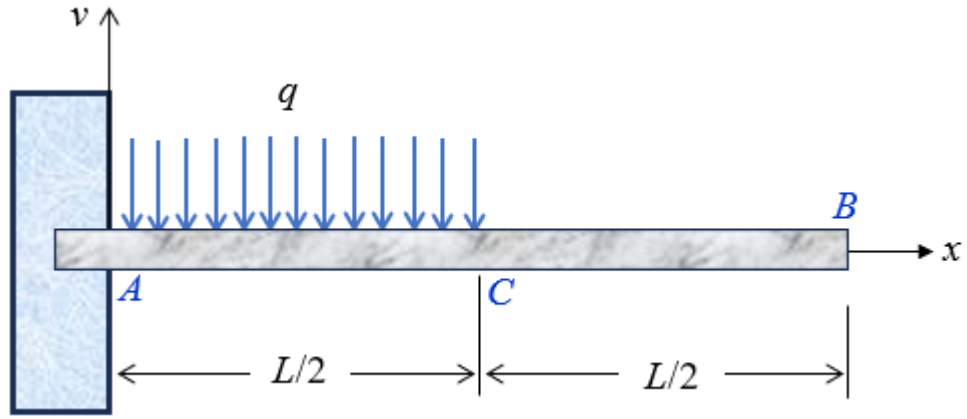


Fig. 2a. Applied load on released structure

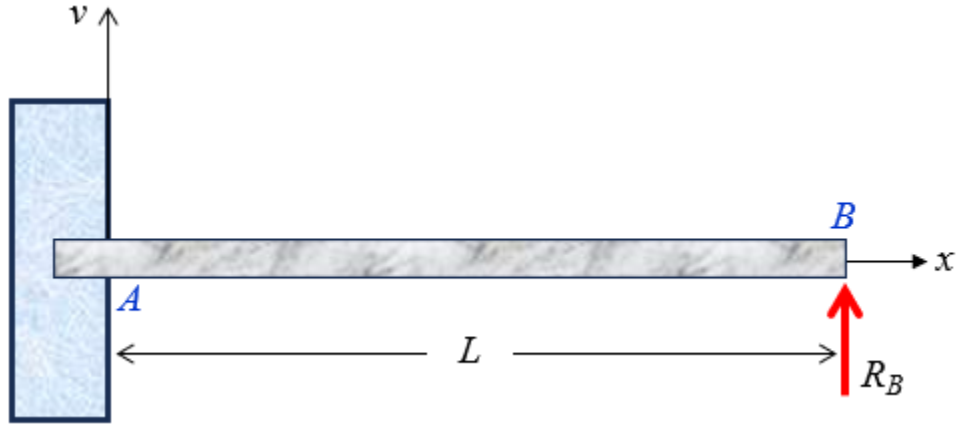


Fig. 2b. Redundant reaction on released structure

Solution

Select the subproblems: We are to select subproblems from the cantilever beam table. This means that R_B is to be considered the redundant reaction. The constituent subproblems are shown in Fig. 2.